

The UX of Administrative Justice: Standardizing Citizen-Facing AI Decision Receipts in Public Sector Automation

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Abstract

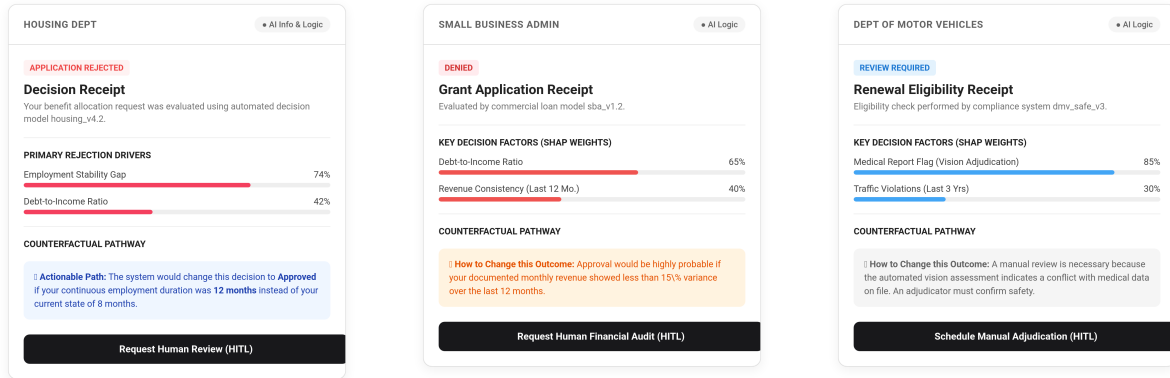
As public administrative agencies increasingly deploy automated decision-making (ADM) systems to manage essential civic pipelines, the tension between algorithmic efficiency and administrative justice has intensified. While overarching legislation like the European Union (EU) AI Act and the General Data Protection Regulation (GDPR) establish a theoretical “Right to Explanation” and mandate human oversight for high-risk systems, a profound operational gap remains. Current public-sector transparency efforts are heavily fragmented, manifesting either as abstract, high-level municipal algorithm registries or as hyper-technical developer-facing model cards. Neither format provides immediate, accessible, or individualized recourse to an ordinary citizen who has just been denied a public service, housing voucher, or business license by an automated system. This paper introduces a novel, citizen-centric framework called the AI Decision Receipt to bridge the missing link between high-level legal theory and practical user experience (UX) design in civic technology.

The proposed AI Decision Receipt standardizes individual algorithmic accountability by translating complex machine learning outputs into a highly scannable, everyday disclosure artifact delivered at the exact moment of an administrative rejection. The real-world adaptability and utility of this framework are evaluated across three distinct public sector domains, which are synthesized comprehensively in Figure 1: AI Label Receipt for Government automated decision-making. The first case study models a high-stakes welfare decision within a municipal Housing Department, illustrating a binary application rejection driven by temporal economic factors. The second case shifts to a financial paradigm, capturing a Small Business Administration grant denial where local explainability metrics are explicitly displayed. The third case evaluates a complex regulatory scenario from the Department of Motor Vehicles, wherein an eligibility renewal is flagged for conditional human intervention due to a data conflict rather than a flat rejection. Together, these three models demonstrate how variable algorithmic parameters can be uniformly harmonized into an intuitive, consistent interface for the public.

Architecturally, the receipt framework relies on three core citizen-facing data features to ensure complete transparency and data accuracy. First, an Input Snapshot visibly mirrors the exact personal parameters ingested by the model, enabling citizens to instantly audit data provenance and catch entry errors or corrupted legacy database flags. Second, a visualized Weight and Impact Breakdown leverages Explainable AI (XAI) pipelines, such as SHAP or LIME, to illustrate which factors most heavily influenced the negative decision. Third, the receipt generates a Counterfactual Pathway, which provides a clear, natural-language explanation outlining the minimum criteria required to change the machine’s outcome. By shifting the objective of transparency from global system literacy to hyper-local citizen literacy, this triptych of features democratizes the audit process, ensuring that applicants do not require a computer science or law degree to comprehend their administrative standing.

Beyond providing passive explanation, the AI Decision Receipt serves as an active, legally enforceable vehicle for individual redress through an embedded, one-click Human-in-the-Loop (HITL) escalation mechanism. As shown in the composite cases of Figure 1, the receipt

transforms abstract statutory protections into a functional escape hatch, allowing citizens to immediately appeal automated findings, freeze their administrative data state, and route their file directly to a human caseworker under mandated service level agreements. To protect the integrity of this process, the framework introduces a backend Blind Review Protocol designed to mitigate automation bias, forcing human caseworkers to evaluate raw application data independently before reviewing the model’s scores. Furthermore, the paper addresses systemic counterarguments regarding the “gaming” of public systems by demonstrating that disclosing personal feature weights enforces policy compliance rather than exposing proprietary source code. Ultimately, this framework operationalizes the EU AI Act’s core principles of human agency, transparency, fairness, and accountability, establishing a scalable blueprint for the next generation of digital administrative justice.



(a) Housing Department UI

(b) Small Business Admin UI

(c) Dept of Motor Vehicles UI

Figure 1: AI Label Receipt for Government Automated Decision-Making. This composite figure shows the standardized architecture across distinct municipal, commercial, and regulatory operational fields.

As outlined within our core conceptual framework, Figure 1 represents the unified implementation matrix. By presenting cases side-by-side, we demonstrate that regardless of whether an engine leverages localized weights for welfare thresholds (Figure 1a), explicit business risk equations (Figure 1b), or medical verification logic barriers (Figure 1c), the user-facing receipt design remains static, structured, and instantly actionable for the citizen.

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